

Deep Learning

- What is Deep Learning?
 - Differences Between Machine Learning and Deep Learning
 - Applications of Deep Learning
 - Overview of Deep Learning Frameworks (TensorFlow, PyTorch, Keras)
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2. Mathematical Foundations

- Linear Algebra:
 - Scalars, Vectors, Matrices, and Tensors
 - Matrix Operations (Addition, Multiplication, Transpose)
 - Eigenvalues and Eigenvectors
 - Calculus:
 - Derivatives and Partial Derivatives
 - Gradient Descent and Optimization
 - Probability and Statistics:
 - Distributions (Normal, Bernoulli, etc.)
 - Expectation, Variance, and Covariance
 - Bayesian Probability
 - Numerical Computation:
 - Floating-Point Arithmetic
 - Gradient Stability (Vanishing and Exploding Gradients)
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3. Neural Networks

- Basics of Neural Networks:
 - Neurons and Perceptrons
 - Activation Functions (Sigmoid, Tanh, ReLU, Leaky ReLU, Softmax)
- Architecture of Neural Networks:
 - Input, Hidden, and Output Layers
 - Forward Propagation
 - Loss Functions (MSE, Cross-Entropy)
- Training Neural Networks:
 - Backpropagation
 - Optimizers (SGD, Adam, RMSProp)
 - Batch Processing (Batch, Mini-Batch, Stochastic)
- Evaluating Neural Networks:

- Metrics (Accuracy, Precision, Recall, F1 Score, ROC-AUC)
 - Train, Validation, and Test Splits
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4. Deep Learning Architectures

4.1 Feedforward Neural Networks (FNN)

- Implementing FNNs
- Hyperparameter Tuning

4.2 Convolutional Neural Networks (CNN)

- Basics of CNNs:
 - Convolutional Layers
 - Pooling Layers (Max, Average)
- Architectures:
 - LeNet, AlexNet, VGG, ResNet, Inception
- Applications:
 - Image Classification
 - Object Detection and Segmentation

4.3 Recurrent Neural Networks (RNN)

- Basics of RNNs:
 - Sequence Modeling
 - Backpropagation Through Time (BPTT)
- Advanced RNNs:
 - Long Short-Term Memory (LSTM)
 - Gated Recurrent Units (GRU)
- Applications:
 - Time-Series Prediction
 - Natural Language Processing (NLP)

4.4 Transformers

- Attention Mechanism
- Architecture of Transformers
- Applications:
 - BERT, GPT, T5
 - Language Translation
 - Text Summarization

4.5 Generative Models

- Generative Adversarial Networks (GANs):
 - Architecture (Generator and Discriminator)
 - Applications (Image Generation, Style Transfer)
 - Variational Autoencoders (VAEs):
 - Theory and Implementation
 - Applications (Data Augmentation, Anomaly Detection)
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5. Optimization Techniques

- Loss Functions for Regression and Classification
 - Gradient Descent Variants:
 - Momentum, Nesterov
 - Adam, AdaGrad, AdaDelta
 - Learning Rate Schedulers
 - Regularization Techniques:
 - L1/L2 Regularization
 - Dropout, Batch Normalization
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6. Deep Learning Tools and Frameworks

- Introduction to PyTorch:
 - Tensor Operations
 - Building and Training Models
 - Introduction to TensorFlow/Keras:
 - Sequential and Functional APIs
 - TensorFlow Datasets (TFDS)
 - Comparison of PyTorch and TensorFlow
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7. Data Preparation and Augmentation

- Handling Large Datasets
- Data Preprocessing:
 - Normalization and Standardization
 - Handling Missing Data
- Data Augmentation Techniques:

- Image Augmentation
 - Text Augmentation
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8. Advanced Topics

8.1 Transfer Learning

- Pretrained Models (ResNet, Inception, etc.)
- Fine-Tuning and Feature Extraction

8.2 Autoencoders

- Basic Autoencoders
- Denoising Autoencoders
- Applications (Data Compression, Anomaly Detection)

8.3 Reinforcement Learning

- Basics of Reinforcement Learning
- Deep Q-Learning
- Policy Gradient Methods

8.4 Explainable AI (XAI)

- Interpreting Neural Networks
 - Techniques:
 - SHAP, LIME
 - Grad-CAM for CNNs
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9. Model Deployment

- Exporting Models
 - Deployment Tools (TensorFlow Serving, TorchServe)
 - Building APIs for Models (Flask, FastAPI)
 - Optimizing for Mobile and Edge Devices (ONNX, TensorFlow Lite)
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10. Case Studies and Projects

Beginner Projects

- Predicting House Prices with FNNs
- Handwritten Digit Recognition (MNIST)

Intermediate Projects

- Building an Image Classifier (CIFAR-10)
- Sentiment Analysis with RNNs

Advanced Projects

- Developing a GAN for Image Synthesis
 - Implementing a Transformer for Text Summarization
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11. Trends and Research in Deep Learning

- Recent Advancements (Diffusion Models, Stable Diffusion)
- Ethical Considerations in AI
- Exploring Research Papers
- Participating in AI Challenges (Kaggle, Alcrowd)